Problem 3: The Confined Professor

SLG

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English adapted version. This document is an English adaptation of our original French report for the TFJM^2 2022 competition (Problem 3, *Professeur confiné*). It is our own work, by Tancrède Amiel and Luigi de Alfaro.

Abstract

This document addresses Exercise 3 of the TFJM^2 2022 competition, "The Confined Professor". Questions 1, 2 and 3.1 have been fully treated, Question 3.2(a) has been almost entirely studied, and Questions 4 and 5 have been partially treated. Our solutions often follow from a gradual process that led us to refine our first answers. Indeed, at the beginning of our work we put forward certain ideas about the strategies that Timothé had to implement in each situation, but the more we worked on the problem, the more we refined our methods and strategies, which led us to change our initial answers so that the final result would be more efficient and more satisfying. Several of our answers therefore resemble each other, since we found methods that apply to the conditions of several questions, or that we could easily adapt.

Question 1 allowed us to understand the stakes of the problem, by discovering how we would have to think and reason about the questions. Question 2 offered us the opportunity to broaden the ideas we had for Question 1, and was also the setting of discoveries that were decisive in improving our answers to the latter.

Question 3, the last question we fully treated, forced us to confront several of our ideas with the goal of finding the most efficient solution, as we did in Question 3.2.b). The specific condition of this question made the problem more interesting than ever, since it moved us far away from the initial situation and forced us to reinvent our strategies.

Question 4 was partially treated, and Question 5 not at all.

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Problem statement

Source note. The statement below is a summary of the official TFJM² 2022 problem set (Problem 3, *Professeur confiné*), available at tfjm.org. It is not the verbatim official wording and must be replaced by the official text before submission.

Timothé (T) falls ill and asks his colleague Perrine (P) to take over his class for a week. The class has n students and the room has n seats. Perrine seats the students as she wishes. When Timothé returns, he wants to recover the seating plan chosen by Perrine, by asking her questions which she answers at the end of the week, and he wants the list of questions to be as short as possible.

1. Students and seats are numbered from 1 to n. Can Timothé always determine the seating plan? If so, give the minimal number of questions he needs if he may only ask:
   a) "Where was student i sitting?";
   b) closed questions, answered yes or no;
   c) "Was student i sitting in seat p?";
   d) "Which students were sitting from seat p_1 to seat p_k?", where k is fixed by Perrine;
   e) "How many students are correctly placed on the seating plan that I propose?"
2. The same questions as in Question 1, except that Timothé telephones Perrine and receives each answer right after asking his question.
3. The same questions as in the previous questions, except that the class is in a room with $m > n$ seats, which Timothé knows, with still n students.
4. Perrine may lie up to l times, where $l \in \mathbb{N}^*$ is known to Timothé. Redo the previous questions in this setting.
5. Propose and explore other research directions.

0 Preliminaries

Some definitions:

Definition 1 (Logarithm in base a)**.** Let $a \neq 1$ be a real number. For a strictly positive real number x, we define $\log_a(x)$ by the following equality:

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}.$$

Definition 2 (Cardinality of a set)**.** The cardinality of a set is defined as the number of elements belonging to that set. We denote the cardinality of a set A by $\text{card}(A)$.

Definition 3 (Integer part of a number)**.** The integer part of a number x is defined as the integer closest to x and less than x. We denote it $\lfloor x \rfloor$.

Definition 4 (Fractional part of a number)**.** The fractional part of a number x is defined as the difference between this number and its integer part. We denote it here d_x, and $0 \leq d_x < 1$.

Definition 5 (Notation)**.** We denote Timothé by the letter T and Perrine by the letter P.

Definition 6 (Factorial of n)**.** We define $n! = n \times (n-1) \times (n-2) \times (n-3) \times \cdots \times 2 \times 1$.

Definition 7 (Discrete sum)**.** Let $\sum_{i=1}^{n} i = 1 + 2 + \cdots + n$ be the sum of all integers from 1 to n inclusive. We write $\sum_{i=1}^{n} i = \dfrac{n(n+1)}{2}$.

For Questions 1, 2 and the part of 4 written here, we consider a class of n students, $n \in \mathbb{N}^*$, having n seats. For Question 3, we now consider that the class is situated in a room composed of $m > n$ seats, where $m \in \mathbb{N}^* \setminus \{1\}$, and that there are always n students in the class.

1 Question 1

1.1 a) Where was student i sitting?

Theorem 1.1. *For a class of n people occupying n seats, T needs at most $n-1$ questions to determine the seating plan.*

Remark 1.1. Throughout the exercise, we look for the number of questions corresponding to the worst case of the best strategy. To determine the seating plan with certainty, T must consider the worst case, and in that case we seek to optimise this worst case by means of a good strategy.

Proof. If T successively asks the questions "Where was student i sitting?" for all $i \in [\![1; n-1]\!]$, then T will know $n-1$ places corresponding to $n-1$ students. From the reading of P's (Perrine's) answers to this questionnaire, it can identify the unique student not cited in P's answers, who sits in the unique seat not cited in P's answers. It will thus have found the seating plan in $n-1$ questions. □

1.2 b) Closed questions

T asks questions about the seating plan to which one can answer yes or no. We start with a simple case: if $n = 2$, it suffices to take one of the 2 possible seating plans and ask whether it is the right one. If the answer is yes, T has its answer; if the answer is no, then the other seating plan is the right one. It will have needed only one question.

Theorem 1.2. *If $n \geq 3$, the maximal number of questions that T must ask is*

$$k = \lfloor \log_2(n!) \rfloor + 1.$$

Proof. This proof is carried out in two steps of analysis and synthesis: first, we shall try to lower-bound the number of questions of a strategy that would be optimal, and then we shall give an example of a question which corresponds to this lower bound and hence to an optimal strategy.

In general, there are $n!$ seating plans (this is the number of permutations of a set of n elements). Among the set E of the $n!$ plans, for each question about the seating plan, a set O of plans is associated with the answer yes, and another set N (different from $\mathbb{N}$, disjoint from O and complementary to O in E) is associated with the answer no. For example, if we ask the question "Is student 1 in seat 1?", then O is the set of all plans in which student 1 is in seat 1, and N is the set of all the other plans. Thus the sum of the cardinalities of N and O is $n!$.

Let $M = n!$. Let P_i be the set of seating plans still possible after question i, $i \in \mathbb{N}^*$. The worst case is that, for each question, the seating plan sought lies in the set whose intersection with P_i has the largest cardinality. Since it is the worst case that interests us, we seek to minimise the larger of the cardinalities $\text{card}(O \cap P)$ and $\text{card}(N \cap P)$.

If M is even, the best first question is therefore one such that $\text{card}(O) = \text{card}(N) = M/2 = \text{card}(E)/2$. After this question, $\text{card}(P_1) = M/2$. Let k be the number of questions in total. For k questions to suffice to find the seating plan, we need $\text{card}(P_k) = 1$.

Suppose there exists $a \in \mathbb{N}^*$ such that $M = 2^a$. We show that the best solution is to divide each P_i by 2 until we reach 1. We solve

$$\frac{M}{2^k} = 1 \iff M = 2^k \iff \ln(M) = k \times \ln(2) \iff k = \log_2(M),$$

since the natural logarithm is a bijection from $\mathbb{R}_+^*$ onto $\mathbb{R}$. Note that k is indeed an integer, since $k = \log_2(M) = \log_2(2^a) = a \in \mathbb{N}^*$.

Now suppose $M = 2^a + 1$. Then after k questions, one possibility remains in addition (otherwise it would be necessary to do better than dividing by 2 at each step, which is impossible), so T must ask one additional question. It follows easily that if $M > 2^a$ then $k > a$, so:

$$M > 2^a \implies k \geq a + 1.$$

Moreover,

$$2^{a+1} > M > 2^a \iff a + 1 > \log_2(M) > a,$$

since the logarithm is strictly increasing on $\mathbb{R}_+^*$. Hence $a + 1 = \lfloor \log_2(M) \rfloor + 1$, and so

$$M > 2^a \implies k \geq \lfloor \log_2(M) \rfloor + 1 \iff k \geq \lfloor \log_2(n!) \rfloor + 1.$$

It remains to find a suitable question. Several dichotomy strategies are possible; we chose the following one. T begins by numbering all the possible seating plans from 1 to M, and asks the following questions: "Is the i-th digit of the number corresponding to the correct seating plan, in my numbering, a 1?", for i varying from 1 to k, where k is the number of digits of M, that is of $n!$, in base 2. A yes means that the digit is a 1, and a no means that it is a 0. It only remains to find this number, which is therefore equal to the number of questions, since a yes means that the digit is a 1 and a no that it is a 0.

It remains to check the number of digits. For $n \geq 3$, since $3 \mid n!$, there exists $a \in \mathbb{N}^*$ such that $2^a < n! < 2^{a+1}$, that is, for every $a \in \mathbb{N}^*$, $n! \neq 2^a$, which is equivalent to

$$a < \log_2(n!) < a + 1.$$

Since $n!$ is written with as many digits in base 2 as 2^{a+1}, namely $a+1$ digits, we have $k = a+1 = \lfloor \log_2(n!) \rfloor + 1$. Since $M > 2^a \implies k \geq \lfloor \log_2(M) \rfloor + 1$, our strategy is optimal. □

1.3 c) Is student i sitting in seat p?

Theorem 1.3. *For a class of n students, T must ask at most*

$$(n-1)^2$$

questions.

Proof. To determine the seating plan chosen by P, T will try to find the seat of each student, one student after another. For the first student, T proposes $n-1$ seats. If among P's answers there is a "yes", the student is in that seat and T moves on to the next step. If there are only "no" answers, the student is in the n-th seat (which was not proposed) and T moves on to the next step. Thus T needed $n-1$ questions. It proceeds in the same way for the first $n-1$ students. Once it has read P's answers, T can place $n-1$ students, which leaves only one seat for one student, which lets it find the seating plan with certainty. So T has asked $n-1$ questions for each of the $n-1$ students, that is $(n-1)^2$ questions in total, by the multiplicative principle.

Let us now justify by contradiction that this step is optimal. If T proposes p seats for a student, with $p \in [\![1; n-2]\!]$, and P answers no p times, then T can no longer deduce where this student is. Indeed, the student could then be in $n-p$ different seats. Now

$$p \leq n-2 \iff 2-n \leq -p \iff 2 \leq n-p,$$

so the student could be in more than two different seats, and T cannot deduce the correct seating plan with certainty.

If T only repeats the operation i times, $i \in [\![1; n-2]\!]$, then T cannot deduce where the last students are. There remain $n-i$ seats for $n-i$ students. Now

$$i \leq n-2 \iff 2-n \leq -i \iff 2 \leq n-i,$$

so there remain more than two seats for more than two students, and T cannot find the correct seating plan with certainty. Hence, with the optimal strategy, T must ask at most $(n-1)^2$ questions. □

1.4 d) Set of students sitting from seat p_1 to seat p_k

Theorem 1.4. *For a class of n students, $n \in \mathbb{N}^*$, and for $1 \leq k \leq n$, we can write $n = q(2k-1)+r$, where q is the quotient and r is the remainder of the Euclidean division of n by $2k-1$, with $0 \leq r < 2k-1$. Then:*

1) *If $k = 1$, it suffices for T to ask at most $n-1$ questions.*

2) *If $k > 1$ and $n \geq 2k-1$ and $r \neq 0$, T must ask at most*

$$K = k \times q + (r-1)$$

questions.

3) *If $k > 1$ and $n \geq 2k-1$ and $r = 0$, T must ask at most*

$$K = k \times q$$

questions.

4) *If $n < 2k-1$, i.e. $q = 0$, it is not possible for T to determine entirely the seating plan chosen by P.*

Proof. **1)** If $k = 1$, this means that the question T will ask P is "Who was the student sitting in seat p_i, $1 \leq i \leq n$?". Thus, by asking this question for $n-1$ seats, T will determine the placement of $n-1$ students, and can even determine by elimination, without asking any further question, the placement of the last student not yet placed. In fact, the question T must ask in this case is equivalent to the question it must ask in 1.a), with the only difference that it interrogates about seats and not about students. Nevertheless, it is the same strategy and the same type of question.

2) Secondly, if $k > 1$ and $n \geq 2k-1$, T can set up an even more efficient strategy than the one just described. If T asks P a first question about the students sitting in seats p_1 to p_k, it will know all the students sitting in these seats, without however knowing their respective seats. However, comparing P's answer with her answer to another question, it could learn about one of the students sitting in seats p_{k+1} to p_{2k-1}. Indeed, if after having asked a question about seats p_1 to p_k, T asks a question about seats p_2 to p_{k+1}, the difference between the first and the second answer of P is that the first includes a student that the second does not include, and vice versa. Now, the difference between the two questions is that the first asks about seat p_1 and not about seat p_{k+1}, and the second asks about seat p_{k+1} and not about seat p_1. Thus, thanks to its two questions, T will be able, by reading the answers, to determine the placement of two students. Following the same scheme and continuing to shift by one seat each time, T will determine two students with each comparison of the answer it receives with the previous one. Thus, after having shifted $k-1$ times, it will have obtained the placements of two students with the first two questions, then of two other students with each new question. It will therefore know the places of $2 + 2(k-2) = 2k-2$ students. However, after having shifted $k-1$ times, it will have asked k questions, about $2k-1$ different students. Indeed, at the k-th question, it asks about seats p_k to p_{2k-1}. There is therefore one student whose seat T does not know. However, it happens that the only seat that is still not filled is seat p_k, because this seat could never be in position to be compared during the successive shifts, since it was present in all the sets of seats on which T questioned P. It also happens that this is the only seat in common between the sets of seats of the first and of the k-th question. Thus, by simply comparing the students cited in the two answers, T will be able to determine the student of seat p_k, since they will be the only student present in both answers. Thus, in k questions, T has determined the places of $2k-1$ students.

By repeating this system $q-1$ times, T will know the places of $(2k-1)(q-1)$ students in $k(q-1)$ questions. It then remains to determine the places of $2k-1+r$ students. For the first $2k-1$ of them, it can follow the system stated above in k questions. However, for $r < 2k-1$, it cannot use it in full. It must therefore ask other questions following on from the k previous ones to know the places of the r remaining students: instead of stopping at the set of places $p_{q(2k-1)-k-1}$ to $p_{q(2k-1)}$, it will continue to the set $p_{q(2k-1)-k-1+(r-1)}$ to p_n. However, since T will know all the places up to $p_{q(2k-1)}$, by shifting by one each time, T will only gain information about one student each time. In $k + (r-1)$ questions it will thus have learned the places of $(2k-1) + r$ students, since by elimination it will have been able to determine the place of the last student, since only one empty seat will remain in the classroom.

3) Thirdly, if $r = 0$, T must ask kq questions. It adopts the same strategy but is obliged to ask the last question. Indeed, at the penultimate question, T cannot yet place the two "middle" students of the last group of $2k-1$ students. These are the students in seats $p_{(q-1)(2k-1)+k-1}$ and $p_{(q-1)(2k-1)+k}$, which are included in all the sets of seats from $(p_{(q-1)(2k-1)+1}$ to $p_{(q-1)(2k-1)+k})$ up to $((p_{(q-1)(2k-1)+k-1}$ to $p_{(q-1)(2k-1)+2k-2})$. So T needs one additional question, that is $kq + r - 1 + 1 = kq$ questions, since $r = 0$.

4) Fourthly, if $n < 2k-1$, the system stated above can no longer be used. Indeed, in that case it is not possible for T to determine the places of all the students, because there would be no pair of sets of seats having exactly one seat in common: it would therefore not be possible to determine the student sitting in seat p_k, as T had to do in the previous case. T can ask for all

the sets of seats from p_i to p_{i+k-1}, with $i \in [\![1; n-k+1]\!]$. Now

$$n \leq 2k-2 \iff k \geq \frac{n+2}{2}.$$

Moreover,

$$i \geq 1 \iff i+k-1 \geq k \iff i+k-1 \geq \frac{n}{2}+1.$$

And

$$-k \leq -\frac{n}{2}+1 \iff n-k+1 \leq \frac{n}{2}.$$

Since $i \leq n-k+1$, we get $i \leq \frac{n}{2}$.

- If n is even, then for all $i \in [\![1; n-k+1]\!]$, $i \leq \frac{n}{2} \leq \frac{n}{2}+1 \leq i+k-1$. Hence, for all $i \in [\![1; n-k+1]\!]$: $p_{n/2}$ and $p_{n/2+1}$ belong to $\{p_{i+m} : m \in [\![0; k-1]\!]\}$.
- If n is odd, then for all $i \in [\![1; n-k+1]\!]$, $i \leq \frac{n+1}{2} \leq \frac{n+3}{2} \leq i+k-1$. Hence, for all $i \in [\![1; n-k+1]\!]$: $p_{(n+1)/2}$ and $p_{(n+3)/2}$ belong to $\{p_{i+m} : m \in [\![0; k-1]\!]\}$.

In all cases, the two middle places are always present in P's answers, which makes them indistinguishable. So T cannot determine the seating plan with certainty. □

1.5 e) How many students are correctly placed on this seating plan?

Theorem 1.5. *For a class of n students, $n \in \mathbb{N}^*$, T must ask at most*

$$(n-1) \times (n-1)!$$

questions.

Proof. Let us arrange the seats from left to right, and let seat p_1 be the right-hand neighbour of seat p_n, as follows:

$$p_1, p_2, \ldots, p_{n-1}, p_n.$$

For any seating plan, there are $n-1$ other seating plans in which each student has the same right-hand neighbour. Indeed, it suffices to shift each student from seat p_i to seat p_{i+1}, except the last student who moves back to the first seat. After n shifts, each student returns to their initial seat. We define a *neighbourhood configuration* as the set of n seating plans sharing the same neighbours. There are therefore $\frac{n!}{n} = (n-1)!$ neighbourhood configurations. For example, for 3 students, 123, 231 and 312 form one neighbourhood configuration, and 213, 321 and 132 form the other. For 4 students, 1234, 4123, 3412 and 2341 form one neighbourhood configuration.

An essential property of these configurations is that the sum of the numbers of correctly placed students over the n plans they contain is equal to n. Indeed, each student is in each seat once and only once, and in particular each student is in their correct seat once and only once. More precisely, if a student is in their correct seat, they are in the wrong seat for the next $n-1$ shifts, and whatever the starting point, they will necessarily pass through their correct seat in $n-1$ shifts. Hence, over the n plans, each student is in their correct seat exactly once, and since there are n students, the sum of correctly placed students is n.

Therefore, T only needs to ask $n-1$ plans per neighbourhood configuration, and can deduce the number of correctly placed students on the last plan. In particular, in the neighbourhood configuration of the correct seating plan, the number of correctly placed students is n on the correct plan and 0 on the other plans of the configuration, which allows T to conclude in every case. It asks $n-1$ questions for each of the $(n-1)!$ neighbourhood configurations. By the multiplicative principle, T must ask at most $(n-1) \times (n-1)!$ questions. □

2 Question 2

2.1 a) Where was student i sitting?

Theorem 2.1. *For a class of n students occupying n seats, T needs at most $n-1$ questions to determine the seating plan.*

Remark 2.1. The fact that in this question T has access to immediate answers does not affect the optimal strategy at all. Indeed, questions of the form "Where is student i sitting?" are independent of one another, i.e. one cannot draw conclusions from previous questions to modify and optimise the following questions. Thus the answer to this question is exactly the same as for Question 1.a).

Proof. The proof is the same as the proof of Theorem 1.1. □

2.2 b) Closed questions

Theorem 2.2. *If $n = 2$, T needs only one question to ask P. If $n \geq 3$, the maximal number of questions that T must ask is*

$$k = \lfloor \log_2(n!) \rfloor + 1.$$

Proof. This theorem is identical to Theorem 1.2. Indeed, the specific conditions of Question 2, namely the immediacy of P's answer, do not affect the strategy that T must put in place to optimise its number of questions. The latter asks questions of the form "Is the i-th digit of the number corresponding to the correct seating plan, in my numbering, a 1?" with i varying from 1 to k, but it could equally ask the same question by asking whether it is a 0, which amounts to the same thing. It therefore only has one question to ask P that has a genuine interest within the framework of the strategy it has set up. Thus, whether or not it knows P's answer to the $(i-1)$-th digit, it has no way to optimise its question: it cannot orient its question knowing that for each digit it is identical. □

2.3 c) Is student i sitting in seat p?

Theorem 2.3. *For a class of n students, the maximal number of questions that T must ask P is*

$$\frac{n(n-1)}{2}.$$

Proof. This strategy is an improved version of the strategy set up in Question 1.c), thanks to the new conditions brought by Question 2. To determine the seating plan chosen by P, T will once again try to find the place of each student step by step, student by student. For the first student, T proposes $n-1$ seats. If among P's answers there is a "yes", the student is in that seat and T moves on to the next step. If there are only "no" answers, the student is in the n-th seat (which was not proposed) and T moves on to the next step. So T needed $n-1$ questions.

It proceeds in the same way for the first $n-1$ students, except that this time P's answers are immediate. This means that T will ask questions for the place of student i, $1 \leq i \leq n$, and it will already know the places of all the students for whom it has already asked questions. Thus, when it questions P about student i, it can ask P only about the seats whose occupant it does not yet know, while still removing one question per student, since the status of the last seat can be determined by elimination.

Thus it will ask at most $n-2$ questions for the second student, $n-3$ for the third, ..., 2 for the $(n-2)$-th and 1 for the $(n-1)$-th. It will not even need to ask a question for the n-th student, since to place them only one unoccupied seat remains.

Thus, summing, the maximal number of questions that T must ask P is the sum of the integers from 1 to $n-1$, that is

$$\sum_{i=1}^{n} i = 1 + 2 + \cdots + (n-1) = \frac{n(n-1)}{2}.$$

□

2.4 d) Set of students sitting from seat p_1 to seat p_k

Theorem 2.4. *For a class of n students, $n \in \mathbb{N}^*$, and for $1 \leq k \leq n$, we can write $n = q(2k-1) + r$, where q is the quotient and r the remainder of the Euclidean division of n by $2k-1$, with $0 \leq r < 2k-1$. Then:*

1) *If $k = 1$, it suffices for T to ask at most $m - 1$ questions.*

2) *If $k > 1$ and $n \geq 2k-1$ and $r \neq 0$, T must ask at most*

$$K = k \times q + (r-1)$$

questions.

3) *If $k > 1$ and $n \geq 2k-1$ and $r = 0$, T must ask at most*

$$K = k \times q$$

questions.

4) *If $n < 2k-1$, i.e. $q = 0$, it is not possible for T to determine entirely the seating plan chosen by P.*

Proof. The proof is exactly the same as that of Theorem 1.4. T cannot do better with immediate answers, since knowing the names of the students immediately does not change its strategy, by symmetry of the roles of the students. □

2.5 e) How many students are correctly placed on this seating plan?

Theorem 2.5. *For a class of n students, $n \in \mathbb{N}^*$, T must ask at most:*

- $\dfrac{n^2}{4} + 1$ *questions if n is even;*
- $\dfrac{n^2+7}{4}$ *questions if n is odd.*

Proof. The strategy now adopted proceeds in three steps.

First step. T proposes an arbitrary seating plan, which requires 1 question.

Second step. The aim of this step is to find a seating plan on which T knows whether two students are correctly placed or misplaced.

Let m be the number of correctly placed students on the first seating plan. If $m = n$ or $m = 0$, this step is finished, but these are not the worst cases. If $m \in [\![1; n-2]\!]$ (m cannot be equal to $n-1$), then T swaps the first and the second students. P can then announce five numbers (except if $m = 1$, in which case the last case is excluded):

- $m+2$: both misplaced students have become correctly placed, which is the best case: T knows the places of two students. If T needs a wrong student (see step 3), it takes the initial configuration, in which both are wrong.
- $m+1$: two misplaced students have become one correctly placed student and one misplaced student. T cannot know which of the two is the correctly placed one, but this is not the worst case, because we have information about a correctly placed student. If T needs a wrong student (see step 3), it takes the initial configuration, in which both are wrong.
- m: two misplaced students have remained misplaced. This is the worst case. If T needs a wrong student (see step 3), it takes either the initial or the final configuration, in which both are wrong.
- $m-1$: at the start there was one correctly placed student and one misplaced student, which have necessarily both become misplaced. T cannot know which of the two was correctly placed, but this is not the worst case, because we have information about a correctly placed student. If T needs a wrong student (see step 3), it takes the final configuration, in which both are wrong.
- $m-2$: at the start there were two correctly placed students, which have both become misplaced. This is one of the best cases: T knows the places of two students. If T needs a wrong student (see step 3), it takes the final configuration, in which both are wrong.

T has used one question in this step.

Third step. T takes the third student. If it knows how to place the first two students correctly, it swaps the third with the fourth. However, this is not the worst case. If it does not know how to place the first two students correctly, it swaps the third student with the first one (which is wrong thanks to the first step). Four results are possible with m_1 (the worst case is $m_1 = 0$) being the number of correctly placed students when the first two students are misplaced. T knows m_1 thanks to the second step:

- $m_1 + 2$: T then knows the places of two students. This is the best case. It repeats the process with the next student (the fourth in the sequence).
- $m_1 + 1$: T then knows that in this seating plan, the new first student or the new third student is correctly placed. It can find out which one, but it will do so after the second answer of the form $m_1 + 1$. It passes to the next student (detailed below).
- m_1: neither of the two new students is correctly placed. T passes to the next student (detailed below).
- $m_1 - 1$: T knows that one of the two students was correctly placed. But thanks to the second step, it knows that the first student was misplaced. It can therefore deduce that the third student was initially correctly placed. It repeats the process with the fourth student.

The two worst cases are therefore those where T obtains m_1 or m_1+1. In that case, after having swapped the third and the first, it swaps the third and the second, then the third and the fourth, then the third and the fifth, and so on, until:

- P has answered once $m_1 + 2$, or twice $m_1 + 1$. With $m_1 + 2$ it can place the two students. The answer $m_1 + 1$ can only arrive twice, when the initial third student is in their correct seat, or when the new third student is correctly placed in the third seat.
- Or until the $(n-1)$-th student. If P has never said $m_1 + 2$ nor $m_1 + 1$, the next step will necessarily give $m_1 + 2$ without needing to ask the question. If P has said $m_1 + 1$ once, the next step will necessarily give $m_1 + 1$ without needing to ask the question.

This process therefore takes at most $n-2$ questions (from the first student to the $(n-1)$-th student, without going through the third).

T takes the first configuration in which it obtained m_1+1. One student is necessarily correctly placed between the initial third student and the new third student. It can then swap the new third student with the second (or the first) student, which it knows to be wrong thanks to the second step. If that student was the correctly placed one, it necessarily becomes wrong, and P's answer is m_1. If it was a wrong one, the initial third student remains correct, and we have swapped two wrong students, so the answer is at least m_1+1. The two cases lead to different answers, so one more question makes it possible to find which student was correctly placed between the initial third student and the new third student. With one additional question, we have placed two students. This step takes at most $n-1$ questions for now.

We then move on to the next student whose place we do not know, and repeat the process, swapping them at most with all the other students we do not know how to place, except one (which can be deduced). This gives $n-3$ questions (we do not need to swap with the two students already placed). Having placed two new students, $n-4$ students remain to be placed. We distinguish cases:

- If n is even, T repeats the process at most until only two students remain to be placed. Thanks to the previous questions it knows how many of these two are correct. If both are correct, it stops; if one is correct, the other is also correct, and it stops; if neither is correct, it does not need to ask the question, it suffices to swap them to finish. The number of questions at this stage, since there are fewer at each iteration of the process, is the sum of an arithmetic sequence of common difference 2, first term 3 and last term $n-1$, with $\frac{n}{2}-1$ terms, whose sum is therefore

$$\frac{(n+2)\left(\frac{n}{2}-1\right)}{2} = \frac{\frac{n^2}{2}-n+n-2}{2} = \frac{n^2}{4}-1.$$

- If n is odd, T repeats the process at most until only three students remain to be placed. Thanks to the previous questions it knows how many of these three are correct. If all three are correct, it stops. Otherwise, one student is correctly placed, but it does not know which one. It can be useful to see the situation as a code whose three digits are 1, 2 and 3, with solution 123. There are $3! = 6$ possibilities: one good one, three with one correct digit, and two completely wrong ones:
  - 123: the solution.
  - 231 and 312: no digit is correct. It suffices to shift each digit (student) one place to the right: we obtain 312 (no correct digit) or 123 (we have found it). If there is no correct digit, one knows without asking the question that shifting once more to the right will necessarily give the correct code. T needs only one question to find the correct seating plan.
  - 132, 213 and 321: one digit is correct. Swap the neighbours of the middle one. We obtain 231, 312 or 123. We are in the first three cases, so at most one more question is needed to conclude.

  The number of questions at this stage, since there are two fewer at each iteration of the process, is the sum of an arithmetic sequence of common difference 2, first term 2 and last term $n-1$, with $\frac{n-1}{2}$ terms, whose sum is therefore

$$\sum_{i=1}^{(n-1)/2}(2i) = 2\times\sum_{i=1}^{(n-1)/2} i = 2\times\frac{\frac{n-1}{2}\left(\frac{n+1}{2}\right)}{2} = \frac{n^2-1}{4}.$$

Conclusion. Adding the three steps, we obtain at most, for the number of questions:

- If n is even:
$$\frac{n^2}{4} - 1 + 1 + 1 = \frac{n^2}{4} + 1.$$
- If n is odd:
$$\frac{n^2 - 1}{4} + 1 + 1 = \frac{n^2 + 7}{4}.$$

□

3 Questions 3

From here on, the room has $m > n$ seats, and there are n students.

3.1 Question 3.1: Non-immediate answers

3.1.1 a) Where was student i sitting?

Theorem 3.1. *For a class of n students sitting in a room with $m > n$ seats, T needs at most n questions.*

Proof. This type of question does not depend on the seats themselves. However, unlike Questions 1 and 2, T cannot exempt itself from asking the last question. Indeed, since there are more seats than students, the seat of the last student cannot be deduced by elimination: T must ask the question for this last student as well. □

3.1.2 b) Closed questions

Theorem 3.2. *For a class of $n \geq 3$ students, sitting in a class with $m > n$ seats, T must ask a maximal number of questions of*

$$k = \left\lfloor \log_2\left(\frac{m!}{(m-n)!}\right) \right\rfloor + 1.$$

Proof. The proof of the validity of this theorem is essentially the same as Proof 1.1. The system of questions remains the same: T will ask, in the same way, questions of the form "Is the i-th digit of the number corresponding to the correct seating plan, in my numbering, a 1?" for i varying from 1 to k.

However, in this case the number of seating plans that T numbers in base 2 in order to perform the dichotomy is no longer $n!$ but $\dfrac{m!}{(m-n)!}$. Indeed, if we do not consider empty seats as strictly identical, there are $m!$ seating plans for this room with m seats. However, if we now consider the empty seats as indistinguishable, there will exist among these $m!$ plans a certain number of identical plans, because they have their empty seats in the same places; moreover, these empty seats being identical, their permutations are not distinguishable. Since n students occupy the seats, $m - n$ seats are empty. Since the permutations of the empty seats must not be counted in the calculation of the number of different seating plans, the number of seating plans is an n-arrangement of the m seats, that is $\dfrac{m!}{(m-n)!}$.

The formula of Theorem 3.3 is therefore exactly the same as that of Theorem 1.2, adapted to the specific conditions of Question 3.1. □

3.1.3 c) Is student i sitting in seat p?

Theorem 3.3. *For a class of n students in a room with $m > n$ seats, T must ask at most*

$$n(m-1)$$

questions.

Proof. For student 1, T asks at most $m-1$ questions to learn their place. For student 2, it asks at most $m-1$ questions, and so on. From student 1 to student n (for the last student it is this time necessary to ask the question), it asks $m-1$ questions each time, so in total, by the multiplicative principle, $n \times (m-1)$ questions. □

3.1.4 d) Set of students sitting from seat p_1 to seat p_k

Theorem 3.4. *For a class of n students, $n \in \mathbb{N}^*$, placed in a room with $m > n$ seats, and for $1 \leq k \leq n$, we can write $m = q(2k-1) + r$, where q is the quotient and r the remainder of the Euclidean division of m by $2k-1$, with $0 \leq r < 2k-1$. Then:*

1) *If $k = 1$, T must ask at most $m-1$ questions.*
2) *If $k > 1$ and $m \geq 2k-1$ and $r \neq 0$, T must ask at most*

$$K = k \times q + (r-1)$$

questions.

3) *If $k > 1$ and $m \geq 2k-1$ and $r = 0$, T must ask at most*

$$K = k \times q$$

questions.

4) *If $m < 2k-1$, i.e. $q = 0$, it is not possible for T to determine entirely the seating plan chosen by P.*

Proof. The proof is exactly the same as that of Theorem 1.4, systematically replacing n by m. T cannot adapt its strategy, since it does not know in advance the empty seats, and in this question P does not answer immediately. As there could be students in every seat, T differentiates all the seats, including the empty ones: it treats the empty seats here as students. □

3.1.5 e) How many students are correctly placed on this seating plan, with a proposed seating plan?

Theorem 3.5. *For a class of n students, $n \in \mathbb{N}^*$, placed in a room with $m > n$ seats, and in order to find the seating plan chosen by P, T must ask at most*

$$\frac{(m-1)!}{(m-n)!} \times (m-1)$$

questions.

Proof. The strategy used to answer this question is the same as that of Question 1.e), but adapted to the specific conditions of Question 3, namely the existence of a number $m-n$ of empty seats in the classroom.

Let us compute the number of neighbourhood configurations in this case. There are $\dfrac{m!}{(m-n)!}$ permutations of the seating plan possible, once one takes into account that the empty seats are

indistinguishable, as explained in Proof 3.2. Now, in a neighbourhood configuration, all students keep their right-hand neighbour at each shift by one seat, and return to their initial seat after m shifts (since T cannot distinguish students from empty seats, having no information about the latter except their number, and is therefore obliged to count them as if they were other students). Hence there exist

$$\frac{m!}{(m-n)!} \times \frac{1}{m} = \frac{(m-1)!}{(m-n)!}$$

neighbourhood configurations.

Moreover, as shown in Proof 1.5, for each neighbourhood configuration it is possible to determine the number of correctly placed students in a given permutation, when one knows the number of correctly placed students in all the other permutations of its neighbourhood configuration. Consequently, T can save itself from asking one additional question per neighbourhood configuration, that is only $m-1$ for each. So, by the multiplicative principle, T must ask at most, a number of questions of

$$\frac{(m-1)!}{(m-n)!} \times (m-1).$$

□

3.2 Question 3.2: Immediate answers

3.2.1 a) Where was student i sitting?

Theorem 3.6. *For a class of n students in a room with $m > n$ seats, T needs at most n questions.*

Proof. The proof is the same as that of Theorem 3.1. □

3.2.2 b) Closed questions

Theorem 3.7. *For a class of $n \geq 3$ students, placed in a room with $m > n$ seats, T must ask at most*

$$k = \left\lfloor \log_2\left(\frac{m!}{(m-n)!}\right) \right\rfloor + 1$$

questions.

Proof. The proof of the validity of this theorem is essentially the same as Proof 1.1. Indeed, adding to Question 3.2 the conditions of Question 2, namely the immediacy of P's answer, changes nothing in the strategy. The system of questions remains the same: T asks, in the same way, questions of the form "Is the i-th digit of the number corresponding to the correct seating plan, in my numbering, a 1?" for i varying from 1 to k. Thus, whatever the answer of P to a question, T cannot adapt its next question, knowing only that the next digit is either a 1 or a 0.

The only difference between this case and that of Question 1.b) is that the number of seating plans that T numbers in base 2 in order to perform the dichotomy is not $n!$ but $\frac{m!}{(m-n)!}$, because if we do not consider empty seats as strictly identical there are $m!$ seating plans for a room with m seats, whereas if we consider the empty seats as indistinguishable there are $\frac{m!}{(m-n)!}$ of them. Indeed, since n students occupy the seats, $m-n$ seats are empty, and their permutations must not be counted in the number of different seating plans; the number of seating plans is an n-arrangement of the m seats, namely $\frac{m!}{(m-n)!}$. The formula of Theorem 3.7 is therefore exactly the same as that of Theorem 1.2, adapted to the specific conditions of Question 3.2.

Although this strategy works perfectly in the case considered, there exists another possible strategy for this question, which would give a maximal number of questions to ask, relatively

close to $k = \left\lfloor \log_2\left(\frac{m!}{(m-n)!}\right)\right\rfloor + 1$. Since the only restriction placed on the questions that T may ask is that the answer must be "yes" or "no", T can begin by determining the $m-n$ empty seats. By asking at most $m-1$ times the question "Is seat i occupied?", for $1 \leq i \leq m$, T determines for each seat whether it is occupied or not. After the $(m-1)$-th question, if T has obtained n answers "yes", it knows that the m-th seat is empty; if it has obtained $n-1$ answers "yes", it knows that the m-th seat will necessarily be occupied. So it does not need to ask a m-th question and can make do with only $m-1$ questions. Moreover, since T would then know the positions of all the occupied seats, it could have adopted the same strategy as in Question 1.b), and in a way act as if the empty seats did not exist. This would give a maximal number of questions for T of

$$k' = \lfloor \log_2(n!) \rfloor + 1 + (m-1) = \lfloor \log_2(n!) \rfloor + m.$$

Proof that $k \leq k'$. Since it is not directly possible to determine which of the two strategies is the best, and gives a maximal number of questions to ask that is lower than the other, we study their difference. We set $K = \log_2\left(\frac{m!}{(m-n)!}\right) + 1$ and $K' = \log_2(n!) + m$, and let d_K and $d_{K'}$ be their respective fractional parts. Thus

$$k - k' = \log_2\left(\frac{m!}{(m-n)!}\right) + 1 - \log_2(n!) - m - d_K + d_{K'}.$$

Hence

$$k - k' = \log_2 \binom{m}{n} + 1 - m - d_K + d_{K'}.$$

If m is even, $\binom{m}{n}$ is maximal when $n = \dfrac{m}{2}$, so that

$$\max(k-k') = \log_2\left(\frac{m!}{\left(\left(\frac{m}{2}\right)!\right)^2}\right) + 1 - m - d_K + d_{K'} = \log_2(m!) - 2\log_2\left(\left(\frac{m}{2}\right)!\right) + 1 - m - d_K + d_{K'}.$$

We now prove by induction that for every even $m \geq 2$ the statement

$$H_m : \quad \log_2(m!) - 2\log_2\left(\left(\tfrac{m}{2}\right)!\right) + 1 - m \leq 0$$

holds.

- **Initialisation.** For $m = 2$: $\log_2(2) - 2\log_2(1!) + 1 - 2 = 1 - 0 + 1 - 2 = 0$, so H_2 holds.
- **Heredity.** Let $m \geq 2$ be even and assume H_m. We must show H_{m+2}, that is

  $$\log_2((m+2)!) - 2\log_2\left(\left(\tfrac{m}{2}+1\right)!\right) + 1 - (m+2) \leq 0.$$

  By H_m, $\log_2(m!) \leq 2\log_2\left(\left(\frac{m}{2}\right)!\right) + m - 1$. Since $(m+2)! = m!\,(m+1)(m+2)$ and $\left(\frac{m}{2}+1\right)! = \left(\frac{m}{2}\right)!\,\left(\frac{m}{2}+1\right)$, we obtain

  $$\log_2((m+2)!) - 2\log_2\left(\left(\tfrac{m}{2}+1\right)!\right) + 1 - (m+2) \leq \log_2(m+1) + \log_2(m+2) - 2\log_2\left(\tfrac{m}{2}+1\right) - 2.$$

  It therefore suffices to show that

  $$\log_2(m+1) + \log_2(m+2) - 2\log_2\left(\tfrac{m}{2}+1\right) - 2 \leq 0.$$

  Let us define, for $x \in \mathbb{R}_+^*$,

  $$f(x) = \log_2(x+1) + \log_2(x+2) - 2\log_2\left(\tfrac{x}{2}+1\right) - 2.$$

The function f is continuous and differentiable on $\mathbb{R}_+^*$, as a sum of usual continuous and differentiable functions on this set. Hence, for all $x \in \mathbb{R}_+^*$,

$$f'(x) = \frac{1}{\ln 2}\left(\frac{1}{x+1} + \frac{1}{x+2} - \frac{2}{x+2}\right) = \frac{1}{\ln 2} \cdot \frac{1}{(x+1)(x+2)} > 0.$$

So f is strictly increasing on $\mathbb{R}_+^*$.

On the other hand,

$$f(x) = \log_2((x+1)(x+2)) - \log_2\left(\left(\tfrac{x}{2}+1\right)^2\right) - 2 \quad \text{and} \quad f(x) = \log_2\left(\frac{x^2+3x+2}{\frac{x^2}{4}+x+1}\right) - 2.$$

Since $x \mapsto \dfrac{x^2+3x+2}{\frac{x^2}{4}+x+1}$ is a rational function,

$$\lim_{x\to+\infty} \frac{x^2+3x+2}{\frac{x^2}{4}+x+1} = \lim_{x\to+\infty} \frac{x^2}{\frac{x^2}{4}} = 4.$$

By continuity of $x \mapsto \log_2(x)$ at $x = 4$, $\lim_{x\to+\infty} \log_2\left(\frac{x^2+3x+2}{\frac{x^2}{4}+x+1}\right) = \log_2(4) = 2$. By sum,

$$\lim_{x\to+\infty} f(x) = 0.$$

Now $f(0) = \log_2(2) - 2 = -1$, and since f is strictly increasing on $\mathbb{R}_+^*$, we have for all $x \in \mathbb{R}_+^*$: $f(x) \leq \lim_{t\to+\infty} f(t)$, that is $f(x) \leq 0$. Hence $\log_2(m+1) + \log_2(m+2) - 2\log_2\left(\frac{m}{2}+1\right) - 2 \leq 0$, and since $m+2$ is also even, H_{m+2} holds.

Conclusion: for all $m \in \mathbb{N}$, $m \geq 2$ and m even, $\log_2(m!) - 2\log_2\left(\left(\frac{m}{2}\right)!\right) + 1 - m \leq 0$. Hence

$$\max(k - k') \leq -d_K + d_{K'}.$$

Since $0 \leq d_{K'} < 1$ and $-1 < -d_K \leq 0$, we have $-1 < d_{K'} - d_K < 1$. Thus $\max(k-k') < 1$, and since $\max(k-k') \in \mathbb{Z}$, we have $\max(k-k') \leq 0$.

The same holds when m is odd. Hence $k \leq k'$ and k is always the best solution. The winning strategy is therefore the one in which T does not determine the empty seats beforehand. □

3.2.3 c) Is student i sitting in seat p?

Theorem 3.8. *For a class of n students in a room with $m > n$ seats, T must ask at most*

$$n \times \frac{2m-n-1}{2}$$

questions.

Remark 3.1. This formula indeed gives an integer: if n is even, then $\dfrac{n}{2} \times (2m-n-1)$ is an integer; and if n is odd, then $n-1$ is even, so $n \times \dfrac{2m-n-1}{2}$ is an integer.

Proof. For student 1, T asks at most $m-1$ questions; indeed, if all the $m-1$ answers are negative, then student 1 is necessarily in the seat on which T has not asked a question. In the same way, T asks $m-2$ questions for student 2, $m-3$ for student 3, and so on, down to $m-n$ questions for student n.

This is easily seen: at that stage $n-1$ seats are occupied, so $n-1$ questions are removed, and there is no need to ask the question for the $(m-(n+1)+1)$-th seat if all answers were

negative before. This gives $m-(n-1)-1=m-n$ questions to ask for the last student. This is again an arithmetic sequence of common difference -1, with first term $m-n$ and last term $m-1$, which gives the maximal number of questions to ask, namely

$$n \times \frac{2m-n-1}{2}.$$

□

3.2.4 d) Set of students sitting from seat p_1 to seat p_k

Theorem 3.9. *For a class of n students, $n \in \mathbb{N}^*$, and for $1 \leq k \leq n$, we can write $m = q(2k-1)+r$, where q is the quotient and r the remainder of the Euclidean division of m by $2k-1$, with $0 \leq r < 2k-1$. Then:*

1) *If $k=1$, T must ask at most $m-1$ questions.*
2) *If $k>1$ and $m \geq 2k-1$ and $r \neq 0$, T must ask at most*

$$K = k \times q + (r-1)$$

questions.

3) *If $k>1$ and $m \geq 2k-1$ and $r=0$, T must ask at most*

$$K = k \times q$$

questions.

4) *If $m < 2k-1$, i.e. $q=0$, it is not possible for T to determine entirely the seating plan chosen by P.*

Proof. The strategy that T must use in this case is exactly the same as in Question 3.1.d). Indeed, as said previously, since T cannot know in advance the empty seats, it is therefore obliged to differentiate each of the seats, building up the seating plan little by little by the method stated in Question 1.d). Moreover, the fact that P's immediate answer does not change the number of questions to ask in the worst case is explained in Proof 2.4. □

Remark 3.2. Here, the immediate answer of P allows T to optimise its strategy in some cases, without really changing the number of questions to ask in the worst case. Indeed, on the one hand, since T knows the number n of students and the number $m-n$ of empty seats, if T uses the method described in Question 1.d) but, before having shifted to the set of seats p_{n-k+1} to p_n, it has already managed to find the seats of each of the n students, it can easily deduce that all the seats on which it has not questioned P are empty, which would save it a possibly substantial number of questions.

We could also imagine a case where seats p_1 to p_{ak}, $a \in \mathbb{N}^*$, are empty. In that case, T would only need to ask a questions to know the nature of all these seats: it would begin by asking one question about the set of seats p_1 to p_k; if they are all empty, another about the set p_{k+1} to p_{2k}, ..., until it gets an answer with a non-zero number of non-empty seats. It therefore seems that the worst case here would be that there is at least one student in the seats p_1 to p_k, that there is at least one student in each set of the form p_{1+ak} to $p_{(a+1)k}$, $a \in \mathbb{N}^*$, and that there is a student in seat p_m. This would indeed force T to use all along its series of questions the method stated in Theorem 1.4, since it could not ask questions directly on a set of k seats.

Remark 3.3. We found that an improvement could be made to the theorem. We could modify the strategy by stating that once T has found the places of the first $n-1$ students, it could ask a question about the first set of k seats following the $(n-1)$-th student, and if it finds no student there, it would continue searching by shifting by k seats, and so on, until it finds a set containing the last student. From there, it would perform a dichotomy on this set to find where the n-th student is. The dichotomy would very probably be much more efficient than the original method. However, we cannot manage to determine the worst case specific to this strategy, nor a formula describing, from the variables of the statement, the number of questions needed to carry out this dichotomy. We could also consider carrying out this dichotomy to determine the places of all the students; however, the problem of quantifying the number of questions persists.

Remark 3.4. To give an example, take a room with $m = 13$ seats, $n = 3$ students and $k = 4$, with the following arrangement:

p_1 : empty; p_2 : empty; p_3 : empty; p_4 : student 1; p_5 : empty; p_6 : empty; p_7 : empty; p_8 : empty;

p_9 : student 2; p_{10} : empty; p_{11} : empty; p_{12} : empty; p_{13} : student 3.

One could determine student 1 with the original method in 4 questions, then with 2 additional questions find the place of the second student. Then, with the dichotomy, T would check whether the last student is in seats p_{10} and ..., or p_{12} and p_{13}, by asking about the set of seats p_8 to p_{11}. Depending on the answer, it would ask about seats p_7 to p_{10} or p_9 to p_{12}. Thus, in 2 questions it would have found the place of the last student, and in $4+2+2 = 8$ questions it would have found the seating plan, whereas the original method would have done it in $1\times4+\big(13-1\times(2\times4-1)\big) = 10$ questions.

Moreover, if T performs a dichotomy to find the first student, then after its first question it can ask which students are in seats p_3 to p_6. If the student is in this set, it then asks about seats p_4 to p_7 to conclude. If the student is not in seats p_3 to p_6, it asks about seats p_2 to p_5 to conclude. It has therefore placed its student in 2 questions instead of 4 for the first method, and needs $2+2+2 = 6$ questions to finish. This method seems better, but it is difficult to know its worst case, and to compare it with the worst case of the first method, which does not necessarily occur in the same configuration.

4 Questions 4

4.1 Question 4.1: Non-immediate answers

4.1.1 a) Where was student i sitting?

Theorem 4.1. *For a class of n students, T needs at most $n \times (l+1) - 1$ questions.*

Proof. For each student, T asks $l+1$ questions, except for the last one, for whom it can make do with l questions.

Suppose P lies i times about the student on whom T has asked l questions, with $i \in [\![0; l]\!]$. Then she has also cited other seats i times. Let a student sitting in another seat, cited k times, with $0 \leq k \leq i$. We have asked $l+1$ questions (or l for the second student, and $l+1$ for the first, by symmetry of the roles). P could thus have lied j times about this student, with $j \in [\![0; l-i]\!]$. So we have incompatible answers for the two students, and one must discern the truth from the lie. In total there are $2l+1-i-j$ truthful answers against $k+j$ directly incompatible ones, with $k+j \leq i+j$. Now

$$i+j \leq l \iff 2l+1-i-j \geq l+1.$$

So the truthful answers, which are compatible with each other and incompatible with the false ones, number at least $l+1$, whereas P can lie only l times. Hence T is always able to discern the true from the false, for all students.

Moreover, this strategy is optimal: if T asked l or fewer questions for two or more students, P could for example lie l times by saying that student 1 is in the seat of student 2, and tell the truth l times or fewer about student 2. T could then not know which of the two is in the seat of student 2. □

4.1.2 b) Closed questions

Theorem 4.2.
- *If $n = 2$, one question is needed.*
- *If $n \geq 3$, the maximal number of questions that T must ask is $k = \lfloor \log_2(n!) \rfloor + 1$.*

Proof. The proof is exactly the same as that of Theorem 1.2, except that the question is different. T asks, for i varying from 1 to k, where k is the number of digits of $n!$ in base 2, the question:

> "Is it true that if I asked you whether the i-th digit of the correct seating plan, in my numbering, is a 1, and you answered me with the same truth value as for this question, you would say yes?"

We show that one knows the true answer in every case:

- The i-th digit is indeed a 1.
  - She tells the truth: she would have said yes, so she says yes.
  - She lies: she would have said no, so she lies and says yes.
- The i-th digit is a 0 (not a 1).
  - She tells the truth: she would have said no, so she says no.
  - She lies: she would have said yes, so she lies and says no.

So P always says yes when the digit is a 1 and always says no when it is a 0, so we need exactly as many questions as when she does not lie. □

4.2 Question 4.2: Immediate answers

4.2.1 a) Where was student i sitting?

Theorem 4.3. *For a class of n students, T needs at most $n \times (l+1) - 1$ questions.*

Proof. The proof is exactly the same as that of Theorem 4.1. Indeed, the fact that T learns P's answer immediately changes nothing: it cannot tell from her answer whether she is lying or not, and so it cannot predict whether she will lie at the next question. □

4.2.2 b) Closed questions

Theorem 4.4.
- *If $n = 2$, one question is needed.*
- *If $n \geq 3$, the maximal number of questions that T must ask is $k = \lfloor \log_2(n!) \rfloor + 1$.*

Proof. The proof is identical to that of Theorem 4.2. As explained in Proof 4.3, immediacy of the answer brings nothing: T cannot know whether P is lying, and cannot predict whether the next answer is true or not. □

5 Research directions

5.1 P can answer a given number p of questions randomly (random: either she lies, or she tells the truth).

5.2 A student changes seats periodically, after a given number i of questions, while P speaks and answers T's questions; through these answers she describes the seating plan that was given to her, to be seen when she answers.

5.3 A question f) is added, in which T is only allowed to ask who was the right-hand neighbour of the right-hand neighbour of a student i, $1 \leq i \leq n$, and only for a limited number b of students chosen by P.

5.4 (For a circular seating plan) T asks P who is the neighbour of a student i, $1 \leq i \leq n$. P answers, as she wishes, with the number of the right-hand or left-hand neighbour, without T knowing which one she chooses, and without her being able to change her answer if T asks her the same question again for a number z of students, $1 \leq z \leq n$, chosen by P and known to T.

For Questions 5.3 and 5.4, we are interested in the variables b and z, and in the minimal values for which it is possible for T to find the seating plan.

6 Conclusion

This problem, which accompanied us over the last four months, was very enriching. Through the long hours spent thinking about it, turning it over in every direction, and tying our brains in knots, our mathematical mind grew and we managed to propose an answer to this problem. We tried to give the most complete solutions possible, taking into account the time and length constraints imposed on us. The number of subsections grew with each new question: to give a complete solution to this problem we would have had to answer more than forty questions, which prevented us from treating all the aspects and extensions it offered. We have moreover favoured the quality and completeness of our proposed answers over their number. In doing so, we could not explore the whole of Question 4 (and Question 3.2 to a lesser extent), and we confined ourselves to proposing new conditions for the research directions.